

The data use rules

- Participant-level data (one row per person) stays in the Workbench: no copying out, downloading, or screenshots.
- No count of 1–20 participants may be shared, directly or by calculation (differences, percentages). 0 is allowed. Allele counts under 40 are shown as <40.
- Downloads are monitored; participant-level downloads trigger an egress alert.
- Nothing in a GitHub repository (code, comments, outputs) may contain participant data or counts under 20.
- Notify the program 2 weeks before publishing or presenting results.

Articles at support.researchallofus.org/hc/en-us/articles/number: Data User Code of Conduct 22346176432532; Data and Statistics Dissemination Policy 22346276580372; Egress Alert Policy 4407354684052; Large Download Requests 48941559021716; Publication and Presentation Policy 22346271543828; Researcher FAQ 52861503885460. In this lab, `work/` and `lab_data/` hold person-level files; `results/` holds aggregate outputs, the only candidates for download.

Moving around the terminal

<code>pwd</code>	print the folder you are in
<code>ls -lh</code>	list files with sizes (h = human units)
<code>ls -lt head</code>	newest files first
<code>cd aou-statgen-lab</code>	move into a folder
<code>cd .. / cd ~</code>	up one level / home folder
<code>mkdir -p results</code>	make a folder (no error if it exists)
<code>cp a b / mv a b</code>	copy / move or rename
<code>rm file</code>	delete a file (no trash can – careful)
<code>du -sh lab_data</code>	total size of a folder
<code>df -h .</code>	free disk space on this VM
<code>history tail -20</code>	your last 20 commands

Tab completes file names. Up-arrow recalls earlier commands. Ctrl-C stops a running command.

Looking inside a text file

<code>head -n 3 file.tsv</code>	first 3 lines (always look first)
<code>tail -n 3 file.tsv</code>	last 3 lines
<code>less -S file.tsv</code>	scroll through; q quits; <code>-S</code> no wrap
<code>wc -l file.tsv</code>	count lines (rows + 1 header)
<code>column -t file.tsv head</code>	line up columns for reading
<code>head -1 file.tsv tr '\t' '\n' cat -n</code>	number the columns of the header
<code>cut -f 2,5 file.tsv head</code>	keep columns 2 and 5 (tab-separated)
<code>cut -d, -f 2 file.csv head</code>	same, comma-separated
<code>grep -c 'Female' file.tsv</code>	count lines containing a word
<code>grep -n 'NA' file.tsv head</code>	show matching lines, with line numbers
<code>sort uniq -c</code>	count each distinct value (after cut)
<code>awk -F'\t' 'NR>1{s+=\$2} END{print s/(NR-1)}' f.tsv</code>	mean of column 2, skipping the header
<code>zcat file.gz head</code>	look inside a gzipped file

Example: how many rows per sex in the phenotype file:

`cut -f 5 work/aou_pheno.tsv | sort | uniq -c`. Output stays on your screen; it is not a file to share.

Running the lab scripts

<code>bash scripts/00_setup.sh</code>	check tools and settings
<code>less scripts/02_build_phenotype.sh</code>	read a script before running it
<code>PHENO=1d1 bash scripts/02_build_phenotype.sh</code>	one-off setting change
<code>DRY_RUN=1 bash scripts/02_build_phenotype.sh</code>	print the SQL, run nothing
<code>bash scripts/02_build_phenotype.sh list</code>	list available phenotypes
<code>nano config.sh</code>	edit settings (Ctrl-O save, Ctrl-X exit)
<code>source config.sh; echo \$CDR_DATASET</code>	load settings, print one

Settings and environment variables

<code>source config.sh</code>	load the lab settings into this terminal
<code>echo \$CDR_DATASET</code>	v9 CDR: wb-silky-artichoke-2408.C2025Q4R6
<code>echo \$WORKSHOP_BUCKET \$BILLING_PROJECT</code>	bucket and billing project
<code>env grep -i -E 'workspace cdr google'</code>	Workbench variables, if set

The Workbench can define `$CDR_DATASET`, `$WORKSPACE_BUCKET`, `$GOOGLE_PROJECT`, and `$CDR_STORAGE_PATH`, but they are not always set in a new terminal (the “Getting Started” featured-workspace notebooks set them up). The lab’s `config.sh` does not depend on them.

The CDR: BigQuery from the terminal (bq)

<code>bq ls \$CDR_DATASET head -40</code>	list the CDR tables
<code>bq show -schema -format=prettyjson \$CDR_DATASET.person</code>	column names and types of one table
<code>bq query -use_legacy_sql=false -dry_run 'SELECT ...'</code>	report bytes a query would scan; runs nothing, costs nothing
<code>bq query -use_legacy_sql=false -maximum_bytes_billed=25000000000 'SELECT ...'</code>	run with a hard cost cap
<code>bq query ... -format=csv -max_rows=10000000 'SELECT ...' > work/out.csv</code>	save results as CSV (person-level: work/)

Table names go in backticks inside SQL: FROM ``$CDR_DATASET.measurement``. Look up concept IDs in the public [Data Browser](#).

Handy queries (aggregate only)

```
SELECT COUNT(DISTINCT person_id) FROM ` $CDR_DATASET.measurement `
WHERE measurement_source_concept_id = 903133
SELECT concept_id, concept_name, vocabulary_id
FROM ` $CDR_DATASET.concept `
WHERE LOWER(concept_name) LIKE '%height%' LIMIT 20
```

Cloud storage (gcloud storage; gsutil is the older tool)

<code>gcloud storage ls \$WORKSHOP_BUCKET</code>	list a bucket or folder
<code>gcloud storage ls -l gs://.../file</code>	list with size
<code>gcloud storage du -s -r gs://.../folder</code>	total size of a cloud folder (-r readable units)
<code>gcloud storage cat gs://.../file.tsv head -3</code>	peek at a cloud text file without copying it
<code>gcloud storage cp gs://.../file lab_data/</code>	copy cloud → VM disk
<code>gcloud storage cp results/x.png \$WORKSHOP_BUCKET/lab/</code>	copy VM disk → a workspace bucket
<code>gcloud storage cp -r folder \$WORKSHOP_BUCKET/lab/</code>	copy a whole folder
<code>gcloud -billing-project=\$BILLING_PROJECT storage cp ...</code>	requester-pays buckets (e.g., the controlled dataset bucket): name who pays, before storage

Older equivalents: `gsutil ls`, `gsutil cp`, `gsutil -u $BILLING_PROJECT cp ...` (requester pays). An error mentioning “requester pays” or “billing” means the billing flag is missing.

Controlled Tier genomic data (v9)

Root: `gs://vwb-aou-datasets-controlled/v9/`. Below, `wgs/` means `wgs/short_read/snpindel/`. From the v9 Controlled Tier data dictionary; list a folder with `gcloud storage ls` for exact file names. All copies need the billing flag.

<code>wgs/acaf_threshold/plink_bed/</code>	whole-genome variants with allele frequency >1% or allele count >100; PLINK 1, one set per chromosome (chr22 .bed ≈ 250 GB)
<code>wgs/acaf_threshold/pgen/</code> <code>bgen/</code> <code>vcf/</code>	same variants, other formats
<code>wgs/exome/</code> <code>wgs/clinvar/</code>	smaller callsets, same layout
<code>wgs/aux/ancestry/ancestry_preds.tsv</code>	predicted ancestry, 16 PCs per person (research_id = person_id)
<code>wgs/aux/relatedness/</code>	related pairs (kinship > 0.1); samples to drop for an unrelated set
<code>microarray/plink/</code>	genotyping-array data, PLINK 1, all chromosomes in one set

Variant IDs in these .bim files are chr:pos:ref:alt (GRCh38), not rsIDs, and FID is 0. All of Us publishes no HapMap3 subset; see `scripts/prep/make_hm3_subset.sh`.

PLINK 2: looking at genotype files

<code>head -n 2 f.fam</code>	people: FID IID father mother sex pheno
<code>head -n 2 f.bim</code>	variants: chr ID cM position allele1 allele2
<code>wc -l < f.fam; wc -l < f.bim</code>	number of people; number of variants
<code>cut -f 1 f.bim uniq -c</code>	variants per chromosome
<code>plink2 -bfile f -freq -out work/f</code>	allele frequencies → .afreq
<code>plink2 -bfile f -missing -out work/f</code>	missing-call rates per person / variant
<code>plink2 -bfile f -extract ids.txt -make-bed -out work/g</code>	keep only listed variant IDs
<code>plink2 -bfile f -keep ppl.txt -make-bed -out work/g</code>	keep only listed people (FID IID)
<code>plink2 -bfile f -score w.txt 1 2 3 header -out work/s</code>	polygenic score from a weight file
<code>grep -E 'processed skipped' work/s.log</code>	how many weights matched

.bed is binary: never open it with head or less. PLINK writes a .log next to every output; read it.

Git: getting code onto the VM and keeping it current

<code>git clone https://github.com/USER/REPO.git</code>	copy a repository onto the VM
<code>git pull</code>	fetch and apply the latest changes
<code>git status</code>	what changed; what is untracked or ignored
<code>git diff</code>	line-by-line changes not yet committed
<code>git log -oneline head</code>	recent commits
<code>git rev-parse -short HEAD</code>	the commit you are on (record it with results)
<code>git add scripts/x.py</code>	stage a file you intend to commit
<code>git commit -m "Describe the change"</code>	record staged changes
<code>git push</code>	send commits to GitHub (needs a token)
<code>git check-ignore -v config.sh</code>	confirm a file is ignored, and by which rule
<code>git ls-files</code>	list every file git tracks
<code>git restore file</code>	discard uncommitted edits to a file

A .gitignore for Workbench projects

```
config.sh  # filled settings
work/      # person-level files
results/   # outputs (review before sharing)
lab_data/  # copied-in data
.ipynb_checkpoints/
```

Commit code, templates, and documentation only. Before committing a notebook, clear its outputs (JupyterLab: Edit → Clear All Outputs): a saved .ipynb stores its outputs. Check with `git diff -cached` before every `git push`.

GitHub CLI (gh) – on your own computer

<code>gh auth login</code>	sign in once
<code>gh repo create NAME -private -clone</code>	new repository
<code>gh repo clone USER/REPO</code>	clone by short name
<code>gh repo view -web</code>	open the repo in a browser

Getting aggregate results to your computer

1. Run `bash scripts/06_save_run.sh`: the disclosure check must pass.
2. Open each file in `results/` and review it yourself.
3. In the JupyterLab file browser, right-click the file → **Download**.

Never download files from `work/`, `lab_data/`, or the saved archive.